Lesson 3

Mirrors

Objectives

You will be able to:

- Understand how to draw ray diagrams for concave mirrors.
- Understand how to solve problems involving concave mirrors.
- Understand how to draw ray diagrams for convex mirrors.
- Understand how to solve problems involving convex mirrors.

Shape of a Mirror's Surface

- In a **plane mirror**, a parallel beam of light changes its direction.

- In a **concave mirror**, parallel beams of light become a convergent beam, whose rays intersect in the focus of the mirror. Also known as converging mirror.

- In a **convex mirror**, parallel beams become divergent, with the rays appearing to diverge from a common point of intersection "behind" the mirror.





Plane one-way mirror

Convex mirror

Concave mirror





1. Plane Mirrors

1.1 Objects and Images

As you have read many times in this text, everyday terms used in physics often have specific meanings. In the study of geometric optics, two of the most important terms are object and image. The **object** is the item that you see in the absence of any optical devices. You can see an object because light rays are spreading out or diverging from every point on the object.

In most cases, light from a luminous source is shining on the object and the object reflects the light by diffuse reflection. Your eyes collect the diverging rays and focus them on the retinas of your eyes. When an optical device redirects the diverging light rays and makes the rays appear to be coming from a point that is not really on the object, the device has formed an **image**.

An *image* is a representation of an object formed by a mirror or lens. Light from the image is seen. A *real image* is formed when light rays pass through and diverge from the image point. A *virtual image* is formed when the light rays do not pass through the image point but only appear to diverge from that point.

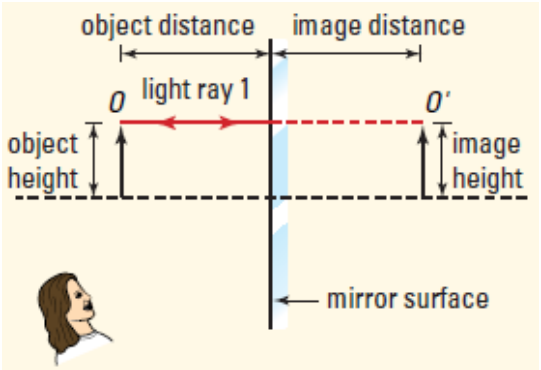
1.2 Drawing Light Ray Diagrams

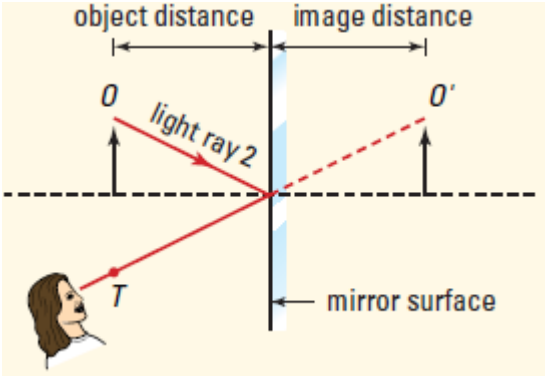
How to draw an image of an object as seen by an eye through a reflection off a flat one-way mirror:

1. Draw a perpendicular line from the object to the mirror.
2. Extend your line an equal distance behind the mirror.
3. Draw a normal halfway between the object and the eyeball.
4. Use the fact that the angle of incidence equals the angle of reflection to draw the rays from object to mirror to eye.
5. Extend the reflected ray.

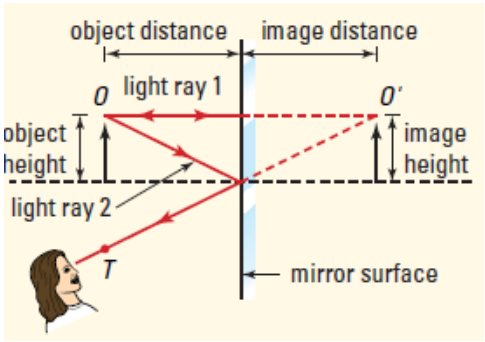
Steps for drawing a plane mirror ray diagram:

- Light ray 1: a ray that strikes perpendicularly to the mirror surface, reflects perpendicularly to the mirror.





- Light ray 2: a ray that strikes the mirror at any angle reflects so that the angle of incidence equals the angle of reflection.
- Extended ray: the image is formed by the extended rays behind the mirror.



In summary, the characteristics of images in plane mirrors are:

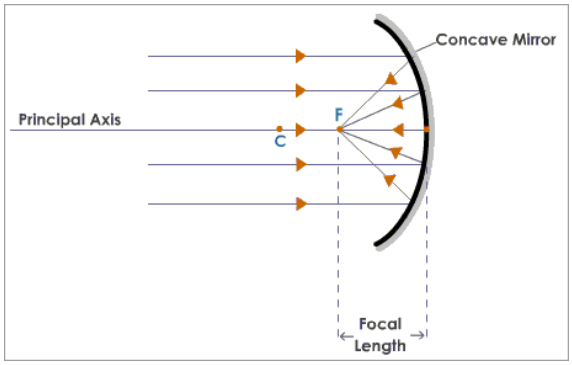
- ☞ The image is the same distance from the mirror as is the object.
- ☞ The image is the same size as the object.
- ☞ The image is always upright.
- ☞ The image is always virtual.

Q: Why does a flat mirror appear to reverse the person looking into a mirror left to right, but not up and down?

A: This means that your image has not been flipped, but actually turned inside out. Think about turning a glove inside out: it's the same thing as the mirror, where your personal image acts as the glove that's being turned front-to-back (for the sake of this lesson, let's neglect the biological and physical impossibilities associated with such a twist!).

2. Concave Mirrors

A concave mirror is a mirror that is rounded inwards. **Concave mirror** is also called a converging mirror. The **center of curvature** (*C*), is located at the center of an imaginary sphere with the same curvature as the mirror. The **radius of curvature** (*R*) is any straight line drawn from the center of curvature to the curved surface. The **focal length** (f) – the distance from the vertex to the focal point, measured along the principal axis. The focal length is related to the radius of curvature by,

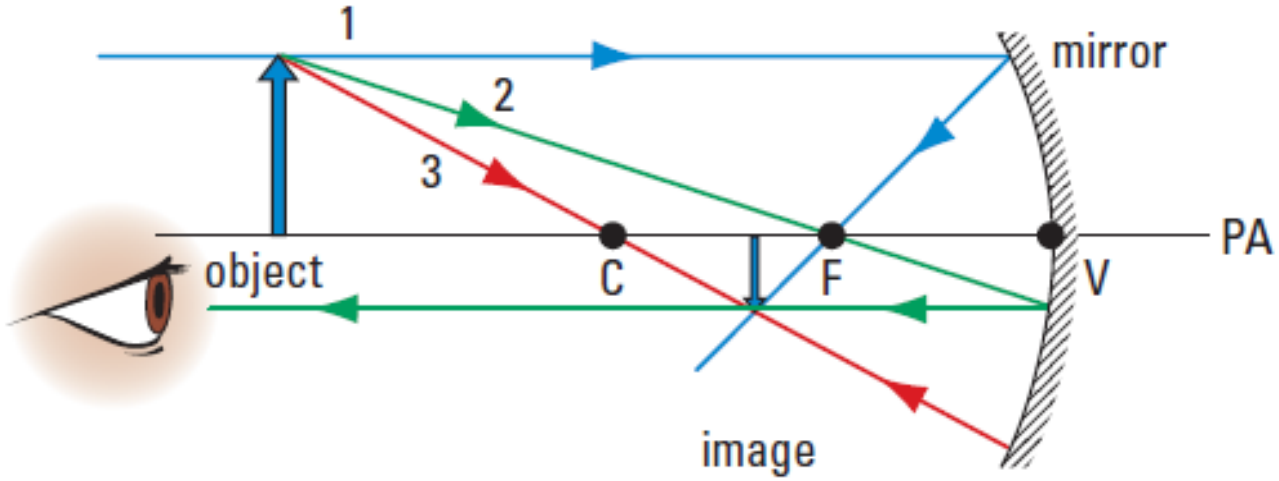


$$f = \frac{R}{2}$$

The geometric center of the actual curved mirror surface is called the **vertex** (*V*). The straight line passing through both the vertex, *V*, and the center of curvature, *C*, is called the **principal axis** (*PA*).

Light Ray Diagrams for Concave Mirrors

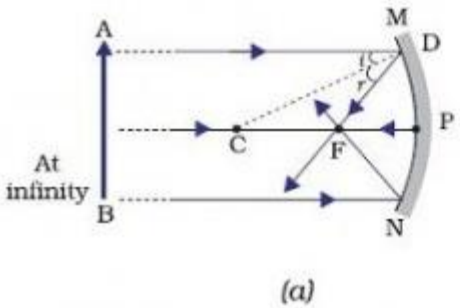
- Light ray 1: a ray that is parallel to the principal axis is reflected through the principal focus (F).
- Light ray 2: a ray passing through the principal focus (F) is reflected parallel to the principal axis.
- Light ray 3: a ray passing through the center (C) is reflected back through the same center point.



- Extended rays if the reflected rays are converging, extend them until they all cross. If the reflected rays are diverging, extend them back behind the mirror with dashed lines.

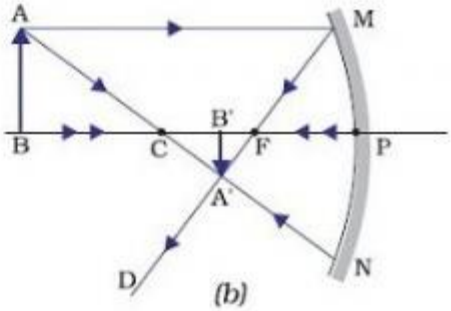
Constructing ray diagrams for concave mirrors:

(a). for object at infinity



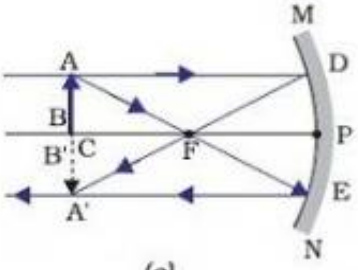
(a)

(b). for object further away than C



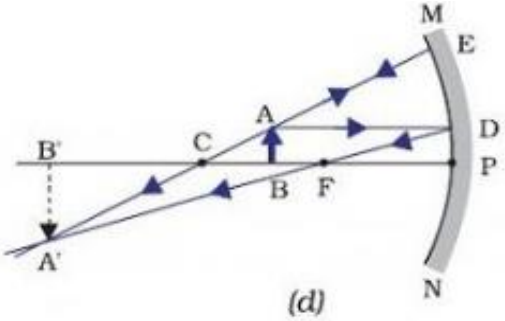
(b)

(c). for object on C



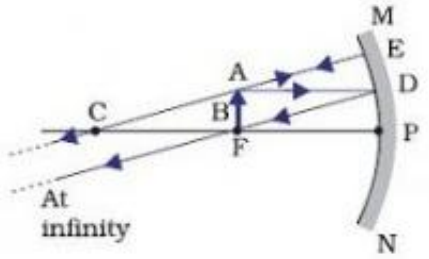
(c)

(d). for object between C and F



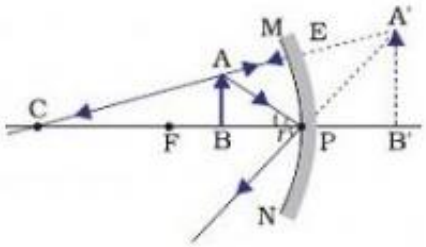
(d)

(e). for object on F



(e)

(f). for object between F and mirror



(f)

	Objection's location or *distance of the object from the mirror*	*Image's location* or *distance of the image from the mirror*
(a)	$d_o = \infty$	$d_i = f$
(b)	$d_o > 2f$	$2f > d_i > f$
(c)	$d_o = 2f$	$d_i = 2f$
(d)	$2f > d_o > f$	$d_i > 2f$
(e)	$d_o = f$	$d_i = \infty$
(f)	$d_o < f$	The image is formed at the back of the mirror.

Equations for Curved Mirrors

For any curved mirror (either concave or convex), the relation between the location of the object and the image formed can be expressed as,

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

where, f is the focal length

d_i is the image distance measured from the mirror

d_o is the object distance measure from the mirror

The size/height of the image is not always the same to the original object (as oppose in the case of flat mirrors). The ratio between the height of the image to the object is known as **magnification**, which is expressed by,

$$m = \frac{h_i}{h_o} = -\frac{d_i}{d_o}$$

It must be stated that the negative sign is used ONLY to *distingquish* whether,

- the image formed is upright ($h_i > 0$) or inverted ($h_i < 0$).

or

- the image is formed in front of ($d_i > 0$) or behind ($d_i < 0$) the curved mirror.

As for d_o and h_o, these two quantities are generally positive.

Sign Conventions for concave/ convex mirrors

	Positive When…	Negative when …
Object's location	Object is in front of mirror	Object is in back of mirror *
Image's location	Image is in front of mirror	Image is in back of mirror
Image's height	Image is upright	Image is inverted
Object's height	Object is upright	Object is inverted *
Focal Length	Mirror is concave	Mirrorn convex

*Generally, not possible!

Exercises

1. An object is located 20cm from a converging mirror that has a focal length of 15cm.
   (a). Where will the image form relative to the mirror vertex?
   (b). If the object is 5 cm high, determine height, and magnification of the image.

Answer:

We know that, $d_o = 20cm$, $f = 15cm$ (converging mirror has a positive focal length), $h_o = 5cm$ (assume that the object is erect)

(a). To locate the image, use the relationship between focal length and object distance, the mirror equation,

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \Rightarrow d_i = \frac{d_0 \times f}{d_0 - f} = 60cm$$

(b). To determine the image height, and magnification, use the equation for magnification,

$$m = \frac{h_i}{h_o} = -\frac{d_i}{d_o} \Rightarrow h_i = -\frac{d_i h_0}{d_0} = -15cm$$

2. An object is placed 15cm from a converging (concave) mirror that has a focal length of 10cm.
   a. Determine the image location.
   b. Determine the magnification factor.
   c. If the object is 5cm high, how high is the image?

3. A concave mirror is designed so that a person 20cm in front of the mirror sees an upright image magnified by a factor of two. What is the radius of curvature of this mirror?

4. If you have a concave mirror whose focal length is 100cm, and you want an image that is upright and 10 times as tall as the object, where should you place the object?

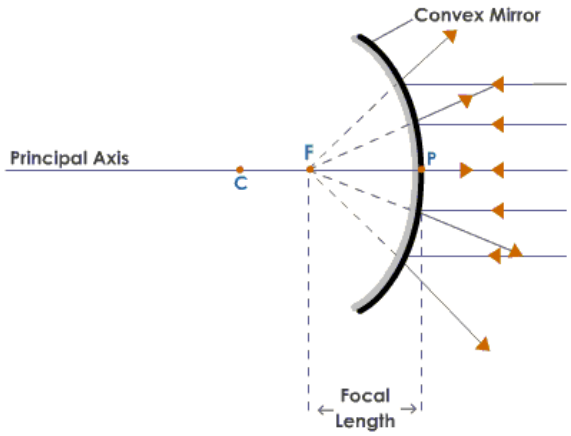
5. A concave mirror has a radius of curvature of 20cm. Locate the image for an object distance of 40cm.

6. A dentist uses a concave mirror to examine a tooth that is 1cm in front of the mirror. The image of the tooth forms 10cm behind the mirror.
   a. What is the mirror's radius of curvature?
   b. What is the magnification of the image?

3. Convex Mirrors

If the outside of the sphere is made into a mirror surface, it is called a **convex mirror**. Convex mirror is also called a diverging mirror as oppose to the concave mirror which is known as converging mirror.

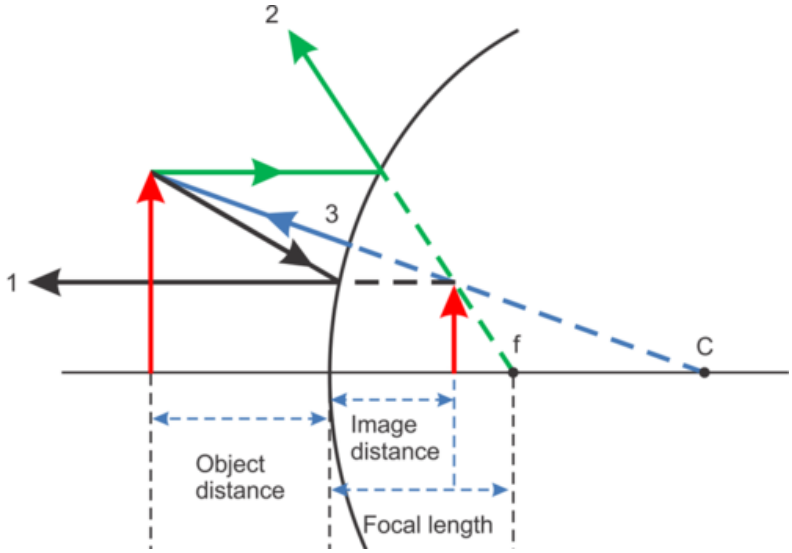
You can determine the properties of an image formed in a convex mirror by using the same light rays as those you used for the concave mirror. Convex mirrors are used when you need to be able to view a large area with just one mirror.



Light Ray Diagrams for Convex Mirrors

- Light ray 1: incident ray parallel to the principal axis appear to come from the principal focus after reflection.
- Light ray 2: a ray appears to pass through the principal focus (F) is reflected parallel to the principal axis.
- Extend the rays behind the mirror until they cross.

For convex mirrors, the image is always virtual, always located inside the mirror, always upright and always smaller than the object.



*Critical thinking: Rear-view mirrors of cars always have a note written "Objects in mirror are closer than they appear." Which curved mirrors you've studied (concave or convex) are used in this example?



1. An automobile rearview mirror shows an image of a truck located 10m from the mirror. The focal length of the mirror is -0.60m.

a. Find the position of the image of the truck.

b. Find the magnification of the image.

Answer: Because the mirror is convex, we expect it to form an upright, reduced, virtual image for any object position.

a. Find the image distance by using Equation

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = -\frac{1}{0.6} - \frac{1}{10}$$

$$d_i = -0.57m$$

b. Find the magnification

$$m = -\frac{d_i}{d_o} = -\frac{-0.57}{10} = +0.057$$

2. Find the image distance of an object placed 5cm in front of a convex mirror whose focal length is 8cm.

3. Find the image distance of an object placed 3cm in front of a convex mirror whose focal length is 8cm.

4. A 4cm tall light bulb is placed a distance 35.5cm from a convex mirror with a focal length of -12.2cm.

a. Find the image distance.

b. Find the image size.

Summary

- For convex mirrors, the reflecting surface is on the outside of the sphere, placing the center of curvature and the focal point on the opposite side of the mirror from the object.
- Light rays that approach the mirror aimed at the focal point will be reflected parallel to the principal axis.
- Light rays that come to the mirror parallel to the principal axis reflect as if they came from the focal point.
- Since the focal point is on the opposite side of the mirror from the object, the focal length is assigned a negative number.
- Both the mirror equation and the magnification equation are true with convex mirrors.